

System design exercise: candidate brief

Version 2.1 · issued 29 September 2026 · replaces version 2.0

You will be asked to design one of the two systems below; the interviewer chooses on the day. The requirements are fixed. Anything not stated here is yours to decide, and you should say your assumptions out loud.

Conventions for all sizing in this brief: a month is 30 days, a day is 86,400 seconds, and storage uses decimal units (1 kB is 1,000 bytes, 1 TB is 1,000 GB).

System A: short-link service

A service that turns a long URL into a short one and redirects visitors who open the short link.

Scale

Requirement	Value
New links created	120 million per month
Read to write ratio	200 redirects for every link created
Peak load	4 times the average, for reads and for writes
Stored size of one link record	500 bytes, indexes included
Click event size	120 bytes per redirect

Service levels

Requirement	Value
Redirect latency	p99 of 80 ms, measured at our edge
Link creation latency	p99 of 300 ms
Availability of redirects	99.99% per month
Availability of the creation API	99.9% per month

Retention

- Links are kept for 5 years from creation, then deleted.
- Click events are kept for 90 days.

Constraints

- Short codes are 7 characters from a 62-character alphabet (a to z, A to Z, 0 to 9).
- Custom aliases are allowed, up to 32 characters.
- A link's destination cannot be edited after creation. A link can be deactivated, and a deactivated link must stop redirecting everywhere within 60 seconds.
- Every new destination is checked against a malware blocklist before the link becomes active.
- Two regions, both serving traffic.
- No more than three storage technologies in the design.

System B: notification system

A platform service that other product teams call to send push notifications, emails and text messages to end users.

Scale

Requirement	Value
Registered users	40 million
Daily active users	9 million
Notifications sent	180 million per day
Channel mix	70% push, 25% email, 5% SMS
Peak load	6 times the average, during campaign bursts
Delivery log entry size	400 bytes

Service levels

Requirement	Value
Transactional notifications	handed to the provider within 2 seconds at p99
Promotional notifications	handed to the provider within 15 minutes
Availability of the send API	99.9% per month

Retention

- Delivery logs: 30 days in hot storage, then 13 months in an archive.
- User preferences: kept for the life of the account.

Constraints

- At most 3 promotional notifications per user per day, across all channels.
- Quiet hours from 22:00 to 08:00 in the user's local time apply to promotional notifications only.
- Transactional notifications are one-time codes and security alerts; they bypass the daily cap and the quiet hours.
- Delivery is at-least-once; callers pass a deduplication key that must be honoured for 24 hours.
- The SMS vendor accepts at most 500 messages per second and charges USD 0.004 per message.
- The push provider accepts at most 20,000 requests per second.

How the session is assessed

Area	Weight
Clarifying requirements and stating assumptions	15%
Capacity estimates	20%
High-level design	30%
Deep dive on one component	25%
Trade-offs and failure handling	10%

Bring your estimates as numbers, not as adjectives. Round sensibly and show the arithmetic.

Changes in version 2.1

- System B added.
- Sizing conventions stated at the top.